# BSIS Program Review

**Ian Alloway**
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## Introduction

Before I started the BSIS program at USF, I knew I liked working with technology and data, but I did not have a real framework for thinking about any of it. I could mess around with code and pull numbers out of a dataset, but I could not have told you much about how information systems are designed, what makes them secure, or how organizations actually use data to make decisions. That changed over the course of this program. The Data Science and Analytics concentration, paired with what I took in intelligence studies, gave me a way to think about all of it that actually stuck.

Through the program I picked up technical skills in database management, Python, R, data analytics, and information security. But the stuff that surprised me was how much the non-technical side mattered. The coursework on ethics, information policy, and organizational behavior forced me to think about the bigger picture around data work, and that has turned out to be just as useful as the coding itself. This essay is a look back at the courses and assignments that shaped me the most, and how they connect to where I am headed next in the MS AI program at USF.

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## Part 1: Overview of Courses

The upper-level LIS courses I took break down into roughly three buckets: core information science foundations, data science and analytics, and intelligence studies.

On the foundations side, LIS 3353, IT Concepts for Information Professionals, was the first course that introduced me to information science as a real discipline. At the time it felt pretty basic, but in hindsight the conceptual vocabulary from that class has come

up more than almost anything else. I use the language from that course constantly when talking about systems, organizations, and where technology fits. LIS 4414, Information Policy and Ethics, ended up being more useful than I thought it would be when I signed up. The legal and ethical frameworks around data privacy, intellectual property, and information governance keep showing up, both in my graduate work and in running Alloway LLC, where I deal with data and user information regularly. LIS 4482, Networks and Communication, filled in a gap I did not realize I had. Understanding how data actually moves through systems has been critical for cybersecurity concepts and for making sense of the infrastructure behind the AI tools I build.

The data science and analytics concentration is where things clicked for me the most. LIS 4800, Introduction to Data Science, was my first real exposure to working with datasets in Python and building out analytical workflows. It was the moment where I went from someone who could write code to someone who could actually do something useful with it analytically. LIS 4805, Data Mining and Predictive Modeling, which I am wrapping up right now, has probably been the hardest and most valuable course in the entire program. We went deep into logistic regression, R programming with object systems like S3 and S4, and data visualization across base R, lattice, and ggplot2. That class pushed me way past the introductory level and gave me real hands-on time with the kind of modeling I want to do as a career. If someone asked me which courses most directly shaped my path toward data science and ML engineering, it would be those two.

The intelligence studies sequence, LIS 4671, 4672, and 4673, covering Introduction to Intelligence Studies, Critical Thinking in Intelligence Analysis, and Open Source Intelligence, taught me something the other courses did not. They taught me to think about information from a security and adversarial angle. Learning the intelligence cycle — which covers planning, collection, processing, analysis, and dissemination — gave me an analytical framework that I have found applies in a lot of places outside the intelligence community. The counterintelligence and OSINT work especially shaped how I think about data integrity and threat modeling, which are topics that keep getting more relevant as AI systems become bigger targets. Those courses also hammered home the importance of ethical reasoning when handling sensitive data, which a purely technical track probably would not have emphasized as much.

If I am being honest, some of the earlier courses in the program covered ground that felt a bit slow given what I already knew going in. But I also recognize that the program

has to work for students coming in with all kinds of backgrounds, and even the slower-paced courses added something to my overall understanding of the field.

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## Part 2: Overview of Key Assignments

Three assignments from the BSIS program stand out more than the rest when I think about what actually moved the needle for me professionally.

The first is the Enhanced Biography and Career Goals paper from the Senior Capstone, LIS 4934. That one forced me to stop and actually write down a clear story about who I am as an information professional. I had to put into words the whole trajectory: relocating from Florida to West Virginia, starting and running Alloway LLC, raising a family while doing all of this, and how the coursework tied it together. The exercise of writing that paper made me realize how much the BSIS program had quietly given me a professional vocabulary. Before the program, I could build things with data. After going through the coursework, I can actually explain why those things matter and where they fit in the bigger picture.

The second is the Case Study Plan, also from the Capstone, where I designed a project to build a centralized software asset database for USF. That assignment was the closest thing to a real-world project management exercise I did in the program. I worked through the six phases of project management: initiation, definition, design, development, implementation, and follow-up. That meant stakeholder identification, requirements gathering, system design, and long-term maintenance planning, all in a realistic institutional setting. It pulled from skills across multiple courses, including database concepts, systems thinking from LIS 3353, and the organizational awareness I built up through the policy and ethics coursework. The biggest takeaway was that technical skills on their own are not enough. You need to understand how organizations work and how to talk to people who are not technical.

The third, and probably the one I am most proud of, is the betwise R package I built for LIS 4805. It is a full R package with 13 exported functions covering Kelly Criterion bet sizing, bankroll simulation, and odds conversion. I wrote unit tests, a vignette, and complete roxygen2 documentation for the whole thing. Building a package from the ground up taught me more about real software engineering practices — like testing, documentation, version control, and package architecture — than any lecture or textbook reading ever did. It also proved to me that I can take an academic concept

and turn it into a polished, functional tool that someone else could actually pick up and use. That is exactly the kind of work I want to be doing professionally: building data-driven tools that solve actual problems.

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## Reflection on Career Benefits

The BSIS program prepared me for my career in ways that go past the specific technical skills. The SQL, Python, and R work is obviously applicable to the data scientist and ML engineer roles I am going after right now. But the less obvious stuff might end up mattering more over time.

The intelligence studies track taught me to approach information systems with a security-first mindset. That is becoming more and more relevant as AI systems turn into targets for adversarial attacks and data manipulation. The ethics and policy coursework gave me a framework for thinking about responsible AI development, which is a topic that keeps coming up in my graduate classes, in the industry, and in the ongoing legislative debates around AI governance.

Running Alloway LLC alongside the coursework was the bridge between the academic side and the real world. The program gave me the concepts and tools. The business gave me the pressure and the context to actually put them to work. As I wrap up the BSIS and move into the MS AI program, I feel good about the foundation. It is solid, and it is going to carry forward no matter what specific technologies or tools I end up working with down the road.

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## Conclusion

Looking back on the BSIS program, it was more than just a collection of courses and grades. It was the process through which I figured out who I am as an information professional. LIS 3353 introduced me to the field. LIS 4805 pushed me into genuinely challenging predictive modeling work. The intelligence studies courses changed how I think about data security and ethics. Each piece of the curriculum contributed something different to how I think and work.

The capstone assignments brought it all together by making me reflect, plan, and build at a level that felt genuinely professional. As I move into the MS AI program and

start my career in earnest, I am not just carrying a degree. I am carrying a way of thinking about data, security, ethics, and human decision-making that I expect will define my work for a long time.